Private Client Graph

Reviewable family and trust relationships from private-client source documents

FTEC5660 hackathon
17 October 2026

Draft for review

The source-review problem

Family and trust facts span notes, letters and later updates.

A graph makes connections visible. A solicitor still needs to inspect the words that support each relationship and distinguish a current appointment from an earlier one.

“Alice Chen confirmed that Bob Chen is a beneficiary of the Evergreen Family Trust.”

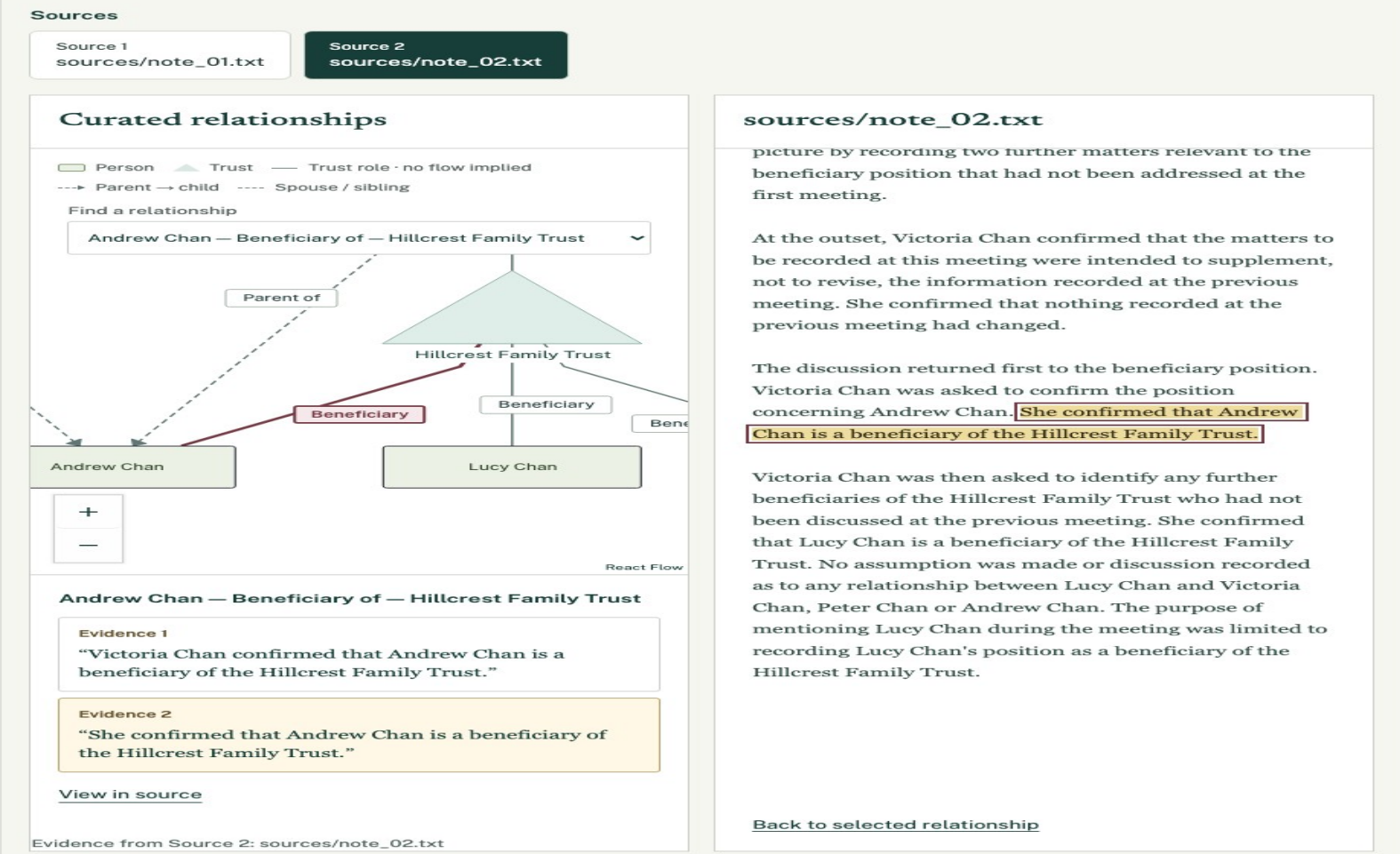
Case 01, synthetic attendance note

Project objective: produce inspectable proposed relationships for professional review.

Evidence beside the relationship graph

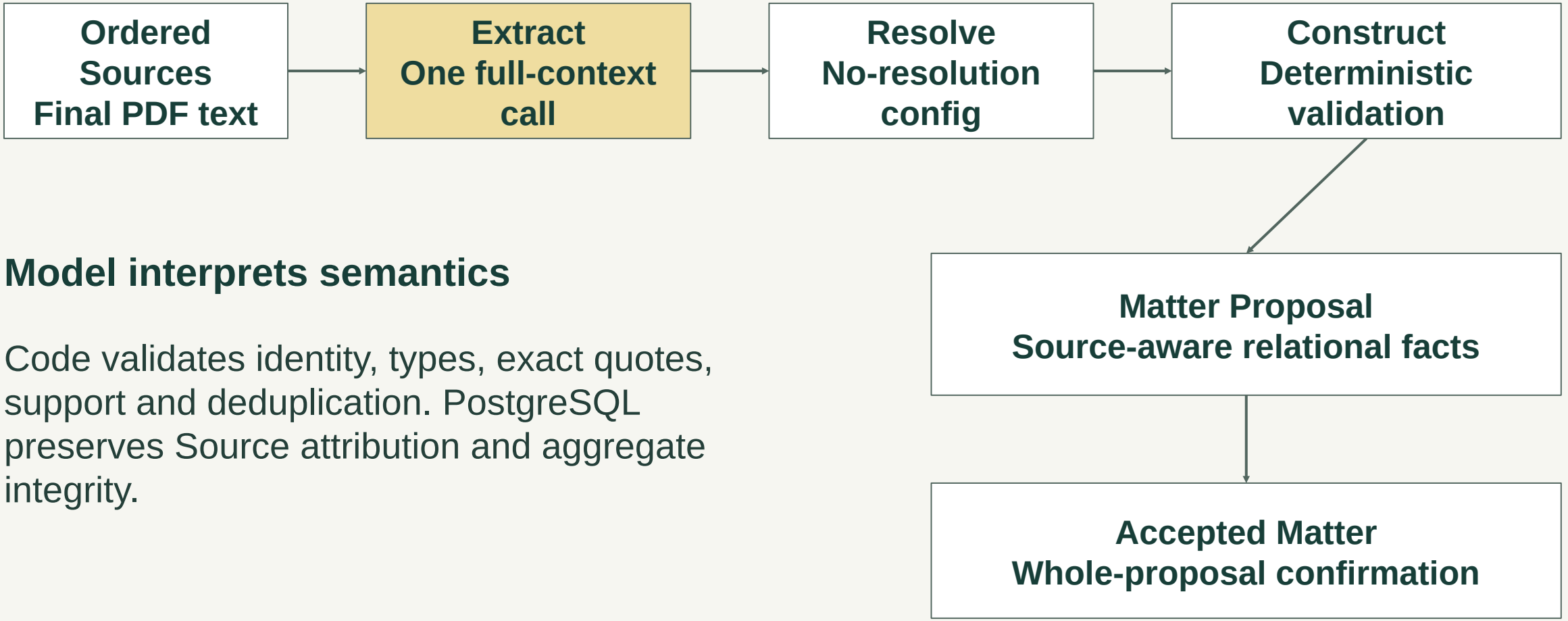
Select a relationship, choose its Evidence and inspect the exact passage in the correct Source.

Practitioners review a saved proposal before confirming the whole graph as current Matter state.



Local public showcase. Curated synthetic Case 04.

System architecture

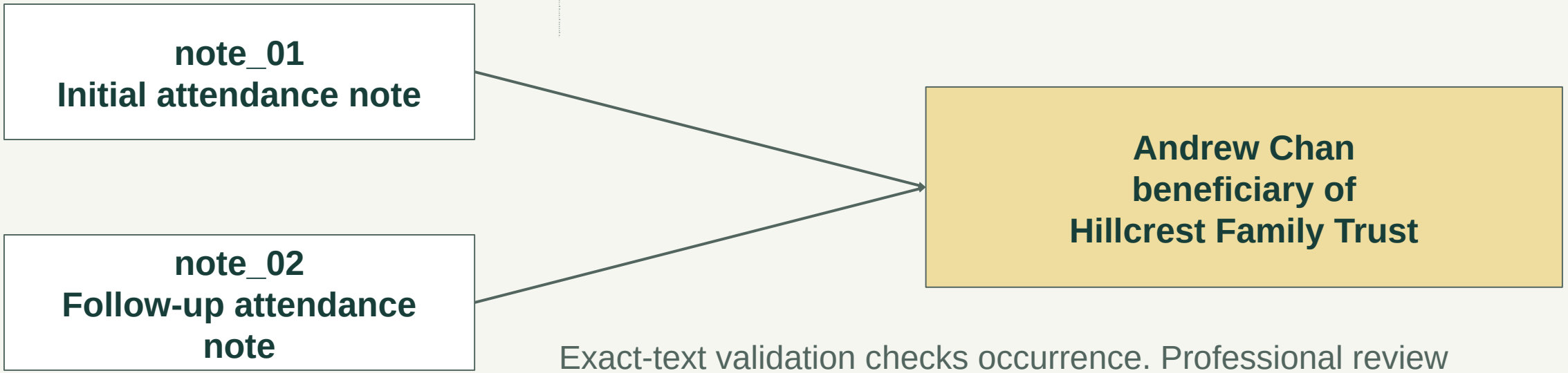


Model interprets semantics

Code validates identity, types, exact quotes, support and deduplication. PostgreSQL preserves Source attribution and aggregate integrity.

Source identity travels with Evidence

A quote carries an explicit Source ID. Repeated titles and repeated words do not identify the document.



Exact-text validation checks occurrence. Professional review checks meaning.

One canonical relationship can retain support from both Sources.

Explicit updates define the current trustee



Replacement effective 10 September

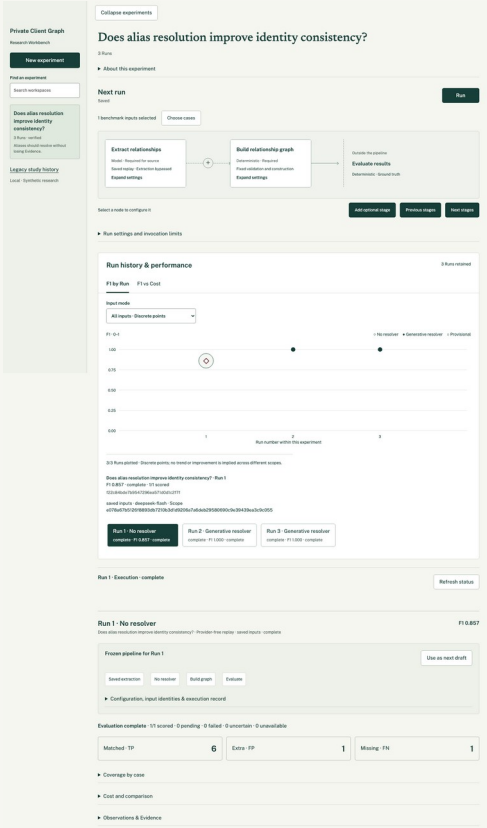
The final current graph retains Rachel’s trustee edge and the five unchanged family and trust relationships.

The source documents retain the historical statement. The application adds no event store or historical-date selector.

Research Workbench and evaluation

Five development cases exercise the chain. Ten separate synthetic cases form the proposed held-out benchmark.

A Run preserves inputs, configuration, extracted candidates, stage outcomes and deterministic evaluation artifacts.



Current Workbench interface. Screenshot context appears in the asset manifest.

Relationship P/R/F1 compare semantic edges. Approved-quote accuracy checks true-positive edges. Source attribution evaluates assessed Evidence items separately.

Draft for review. Evaluation and deployment approval remain pending.

Held-out benchmark status

10 synthetic cases
39 current relationships

Independent fixture review covers exact Source attribution and supported current relationships. Benchmark approval and the frozen-chain evaluation remain pending.

No held-out precision, recall, F1 or provider cost is claimed in this draft.

Findings and limitations

Limits of the current evidence

The held-out examples are small, explicit and synthetic. They test specific difficulties and offer limited evidence of robustness on practice documents.

Exact occurrence and approved-quote scoring cannot establish legal correctness. Curated examples demonstrate review behaviour, not extraction accuracy.

Synthetic material only. Text-layer PDFs only. Whole-proposal confirmation. Real client use needs a separate data-handling and tenancy design.

Conclusion and demonstration

A bounded extraction system with inspectable Evidence and explicit professional confirmation

The next evidence should come from broader documents and practitioner feedback. Real-client ingestion requires a separately approved security and data design.

https://private-client-graph-production.up.railway.app

The public service currently has the earlier Case 01 demo. The reviewed multi-source build remains a local preview pending deployment approval.